THE RIEMANN-HURWITZ FORMULA

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Abstract. These notes were prepared by Yiming Wang for the talk *The Riemann-Hurwitz formula*, given as part of the seminar *Learning algebraic geometry via curves*, organized by Andrea Panontin and Pavel Sechin during the winter semester 2026 – 2027 at the University of Regensburg.

These notes closely follow [Har77, Section 4.2]. The perspective on scheme theory adopted in these notes follows that of [Hoy26]. Throughout these notes, we will work over an algebraically closed field k of characteristic $\neq 2$, a *curve* over k means a smooth, proper, geometrically integral k-scheme of dimension one, and a *point* on a curve means a *closed point* on the curve, unless otherwise stated. Furthermore, the author imposes finiteness assumptions throughout the text; in our setting, this is equivalent to assuming that the morphism is nonconstant.

The author used ChatGPT 5.6 for mathematical and grammatical revision, as well as to fill in several gaps. The document itself has not been directly processed by AI. Any errors are solely the responsibility of the author.

Contents

1. The Riemann-Hurwitz formula 1
2. Examples 3
3. Topological genus 5
References 6

1. The Riemann-Hurwitz formula

Proposition 1.1 (The fundamental exact sequence)**.** *Let $f\colon Y \to X$ be a finite, generically separable morphism of curves over k. There exists a canonical exact sequence*

$$0 \to f^*\Omega_{X/k} \xrightarrow{\mathrm{d}f} \Omega_{Y/k} \to \Omega_f \to 0.$$

Proof. Consider the fundamental exact sequence associated to the morphism $f\colon Y \to X$ over $\mathrm{Spec}(k)$.

$$f^*\Omega_{X/k} \xrightarrow{\mathrm{d}f} \Omega_{Y/k} \to \Omega_f \to 0.$$

Since Y and X are smooth over k, the characterizations of smoothness ([Hoy26, Theorem 9.4]) imply that $\Omega_{Y/k}$ and $f^*\Omega_{X/k}$ are line bundles over Y. By taking the pullback along the generic point, $\eta_Y^*(\Omega_{Y/k})$ is identified with $\Omega_{\kappa(Y)/k}$ (see [Hoy26, Proposition 9.42]). Since $f\colon Y \to X$ is dominant, the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Spec}(\kappa(Y)) & \xrightarrow{\eta_Y} & Y \\ \downarrow & & \downarrow f \\ \mathrm{Spec}(\kappa(X)) & \xrightarrow{\eta_X} & X. \end{array}$$

Therefore, $\eta_Y^* f^*\Omega_{X/k} \simeq \Omega_{\kappa(X)/k} \otimes_{\kappa(X)} \kappa(Y)$. Under these identifications, the morphism $\eta_Y^*\mathrm{d}f$ fits into the fundamental exact sequence induced by $k \subseteq \kappa(X) \subseteq \kappa(Y)$ as follows:

$$\Omega_{\kappa(X)/k} \otimes_{\kappa(X)} \kappa(Y) \to \Omega_{\kappa(Y)/k} \to \Omega_{\kappa(Y)/\kappa(X)} \to 0.$$

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Since $\kappa(Y)/\kappa(X)$ is a finite separable extension of fields, it is étale and $\Omega_{\kappa(Y)/\kappa(X)} = 0$. Therefore, $\eta_Y^* \mathrm{d}f$ is a surjective morphism of 1-dimensional $\kappa(Y)$-vector spaces, hence is an isomorphism.

We will now show that $\mathrm{d}f\colon f^*\Omega_{X/k} \to \Omega_{Y/k}$ is a monomorphism in Mod_Y. Let $(U_i \subseteq Y)_{i\in I}$ be an affine open covering of Y. It suffices to show that $\mathrm{d}f|_U$ is injective. After refining the open cover by principal affine opens, we may assume that $\mathrm{d}f|_U$ is given by multiplication by some element $a \in \Gamma(U, \mathcal{O}_U)$. The induced morphism on fraction fields $a\colon \kappa(Y) \to \kappa(Y)$ is a $\kappa(Y)$-linear isomorphism, hence a is nonzero. For an integral domain, multiplication by a nonzero element is injective, hence $\mathrm{d}f\colon f^*\Omega_{X/k} \to \Omega_{Y/k}$ is a monomorphism. □

Remark 1.2. In Proposition 1.1, the key point is that $\mathrm{d}f$ is nonzero if and only if its restriction to the generic point is nonzero, and this is equivalent to $f\colon Y \to X$ being generically separable.

Definition 1.3 (Ramification divisor)**.** Let $f\colon Y \to X$ be a finite, generically separable morphism of curves over k. Let $L_f = f^*\Omega_{X/k} \otimes \Omega_{Y/k}^\vee$. The effective Cartier divisor $s_f\colon L_f \hookrightarrow \mathcal{O}_Y$ is the *ramification divisor* of f.[1]

For a finite morphism $f\colon Y \to X$ between curves, the *Riemann–Hurwitz formula* relates the genera of Y and X to the ramification of f. We begin by making the notion of ramification precise.

Definition 1.4 (Ramification index)**.** Let $f\colon Y \to X$ be a finite morphism of curves, $y \in Y$ and $x = f(y)$, with the induced morphism on stalks $f_y\colon \mathcal{O}_{X,x} \to \mathcal{O}_{Y,y}$. Then $f_y(t_x) = u t_y^{e_y}$ for some $e_y \in \mathbf{N}_{\geq 1}$,[2] where t_y and t_x are chosen uniformizers of $\mathcal{O}_{Y,y}$ and $\mathcal{O}_{X,x}$, and $u \in \mathcal{O}_{Y,y}^\times$. The number e_y is the *ramification index* of f at y.

Definition 1.5 (Ramification)**.** Let $f\colon Y \to X$ be a finite morphism of curves, $y \in Y$ and $x = f(y)$.

(a) f is *ramified* at y and x is a *branch point* of f, if $e_y > 1$;
(b) f is *unramified* at y, if $e_y = 1$;
(c) the ramification of f at y is *tame*, if $\mathrm{char}(k) = 0$, or $\mathrm{char}(k) = p$ and $p \nmid e_y$;
(d) the ramification of f at y is *wild*, if $\mathrm{char}(k) = p$ and $p \mid e_y$.

From Definition 1.5, we deduce another characterization of unramifiedness:

Remark 1.6. Let $f\colon Y \to X$ be a finite morphism of curves over k, $y \in Y$ and $x = f(y)$. Then f is unramified at y in the sense of Definition 1.5 if and only if it is unramified in the scheme-theoretic sense.

Indeed, the stalk of the quasi-coherent sheaf corresponding to $\Omega_{Y/X}$ at y is computed as $\Omega_{\mathcal{O}_{Y,y}/\mathcal{O}_{X,x}}$. By Nakayama's lemma, it is zero if and only if $\Omega_{\mathcal{O}_{Y,y}/\mathcal{O}_{X,x}} \otimes_{\mathcal{O}_{Y,y}} \kappa(y)$ is zero. By applying $- \otimes_{\mathcal{O}_{Y,y}} \kappa(y)$ to the fundamental exact sequence associated to $k \to \mathcal{O}_{X,x} \to \mathcal{O}_{Y,y}$, we obtain the following exact sequence:

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \otimes_{\kappa(x)} \kappa(y) \xrightarrow{\overline{f}} \mathfrak{m}_y/\mathfrak{m}_y^2 \to \Omega_{\mathcal{O}_{Y,y}/\mathcal{O}_{X,x}} \otimes_{\mathcal{O}_{Y,y}} \kappa(y) \to 0,$$

as $\Omega_{\mathcal{O}_{X,x}/k} \otimes_{\mathcal{O}_{X,x}} \kappa(x) \simeq \Omega_{X/k}(x) \simeq \mathfrak{m}_x/\mathfrak{m}_x^2$ (see [Hoy26, Remark 9.59]). The $\kappa(y)$-linear morphism $\overline{f}$ is given by $\overline{f}([t_x] \otimes 1) = [u t_y]$, whenever $e_y = 1$, and zero otherwise. Therefore, $\mathrm{coker}(\overline{f}) = 0$ if and only if $e_y = 1$.

Remark 1.7 (Cotangent module of a curve)*.* Let Y be a curve over k and $y \in Y$. The cotangent module $\Omega_{\mathcal{O}_{Y,y}/k}$ is a free $\mathcal{O}_{Y,y}$-module of rank 1, generated by $\mathrm{d}t_y$.

Indeed, $\Omega_{Y/k}$ is a line bundle over Y, hence the pullback bundle $\Omega_{\mathcal{O}_{Y,y}/k}$ is a line over the local ring $\mathcal{O}_{Y,y}$, hence is a free of rank 1. Under the canonical isomorphism

$$\mathfrak{m}_y/\mathfrak{m}_y^2 \simeq \Omega_{\mathcal{O}_{Y,y}/k} \otimes_{\mathcal{O}_{Y,y}} \kappa(y),$$

the class of t_y maps to the class of $\mathrm{d}t_y$. Nakayama's lemma shows that $\Omega_{\mathcal{O}_{Y,y}/k}$ is generated by $\mathrm{d}t_y$.

Indeed, the ramification divisor records the ramification in the following precise sense:

Proposition 1.8. *Let $f\colon Y \to X$ be a finite, generically separable morphism of curves over k, $y \in Y$ and $x = f(y) \in X$.*

[1] Classically, the line bundle associated to the effective Cartier divisor $s\colon L \hookrightarrow \mathcal{O}_X$ is $L^\vee$, corresponding to the dual $s^\vee$.
[2] Since $f\colon Y \to X$ is finite, hence dominant, e_y is nonzero.

(a) $\mathrm{mult}_y(R_f) = \mathrm{length}_{\mathcal{O}_{Y,y}}\big((\Omega_f)_y\big)$;
(b) *if f is tamely ramified at y, then* $\mathrm{mult}_y(R_f) = e_y - 1$ *and*
(c) *if f is wildly ramified at y, then* $\mathrm{mult}_y(R_f) > e_y - 1$.
(d) *The support $|R_f|$ is the set of ramification points of f.*

Proof. Let t_y and t_x be uniformizers of $\mathcal{O}_{Y,y}$ and $\mathcal{O}_{X,x}$ respectively.

(a) The fundamental exact sequence associated to $Y \to X \to \mathrm{Spec}(k)$ implies that

$$0 \to \mathcal{O}_{Y,y}\mathrm{d}t_x \xrightarrow{(\mathrm{d}f)_y} \mathcal{O}_{Y,y}\mathrm{d}t_y \to (\Omega_f)_y \to 0$$

is exact (Remark 1.7), where $(\mathrm{d}f)_y$ is induced by f_y, hence is given by $\mathrm{d}t_x \mapsto \mathrm{d}(f_y(t_x)) = c_y\mathrm{d}t_y$ for some $c_y \in \mathcal{O}_{Y,y}$. By the same virtue as Proposition 1.1, c_y is nonzero, which justifies that the first map is injective. Therefore, $(\Omega_f)_y \simeq \mathcal{O}_{Y,y}/(c_y)$. Since the generator $\mathrm{d}t_x \otimes (\mathrm{d}t_y)^\vee$ of $(L_f)_y$ is sent by s_f to c_y, the pullback of R_f to $\mathrm{Spec}(\mathcal{O}_{Y,y})$ is defined by (c_y). Consequently, both sides in (a) are equal to $v_y(c_y)$ ([Hoy26, Construction 10.82]).

(b) Suppose that $f_y(t_x) = ut_y^{e_y}$ for some $u \in \mathcal{O}_{Y,y}^\times$ and $\mathrm{d}u = w\mathrm{d}t_y$ for some $w \in \mathcal{O}_{Y,y}$. By the Leibniz rule,

$$\mathrm{d}\big(f_y(t_x)\big) = \mathrm{d}(ut_y^{e_y}) = wt_y^{e_y}\mathrm{d}t_y + ue_yt_y^{e_y-1}\mathrm{d}t_y = t_y^{e_y-1}(wt_y + ue_y)\mathrm{d}t_y.$$

As f is tamely ramified at y, e_y is invertible in k and $wt_y + ue_y$ is invertible, hence $v_y(c_y) = e_y - 1$.

(c) Since f is wildly ramified at y, by the computation in (b), $c_y = wt_y^{e_y}$, hence $v_y(c_y) \geq e_y > e_y - 1$.

(d) The morphism f is unramified at $y \in Y$ if and only if $(\Omega_f)_y = 0$. By (a), this is the case if and only if c_y is invertible, i.e., (c_y) is not contained in the maximal ideal $\mathfrak{m}_y$, i.e., $y \notin \mathrm{V}(s_f)$.

□

Theorem 1.9 (Riemann-Hurwitz formula)**.** *Let $f\colon Y \to X$ be a finite, generically separable morphism of curves over k. Then the genera of X and Y satisfy*

$$2g(Y) - 2 = n(2g(X) - 2) + \deg(R_f),$$

where $n = [\kappa(Y) : \kappa(X)]$ is the degree of f. Furthermore, if f is tamely ramified*, then* $\deg(R_f) = \sum_{y\in Y}(e_y - 1)$.

Proof. Since tensor products add degrees, whereas dualization negates the degree,

$$\deg(R_f) = -\deg(L_f) = \deg(\Omega_{Y/k}) - \deg(f)\deg(\Omega_{X/k}) = 2g(Y) - 2 - n(2g(X) - 2),$$

where the last equality follows from the computation of the degrees of the canonical divisors of Y and X.

Since Y is of finite type over k and k is algebraically closed, $[\kappa(y) : k] = 1$. Therefore, the second claim follows from Proposition 1.8. □

2. Examples

Definition 2.1 (Hyperelliptic curve)**.** A *hyperelliptic* curve C over k is a curve of genus $g \geq 2$ over k with a finite morphism $\pi\colon C \to \mathbf{P}^1_k$ of degree 2.

Remark 2.2 (Affine charts via normalization)*.* Hyperelliptic curves are given by equations $y^2 = f(x)$, where $f(x)$ is separable over k. Let $\pi\colon C \to \mathbf{P}^1_k$ be a hyperelliptic curve over k. Since π is of degree 2 and $\kappa(\mathbf{P}^1_k) = k(x)$, the function field of C is given by $\kappa(C) = k(x, y)$, where $y^2 = f(x)$ for some separable polynomial $f(x) \in k[x]$.

Since $V_0 = \mathrm{Spec}(k[x])$ and $V_\infty = \mathrm{Spec}(k[t])$ form an open covering of $\mathbf{P}^1_k$, where $t = x^{-1}$, the open subschemes $U_0 = \pi^{-1}(V_0)$ and $U_\infty = \pi^{-1}(V_\infty)$ form an open covering of C.

Furthermore, $U_0 \subseteq C$ is normal, which implies that $\mathcal{O}(U_0)$ is integrally closed in $\kappa(C)$. On the other hand, $\mathrm{Spec}(k[x, y]/(y^2 - f(x)))$ is smooth over k, by the Jacobian criterion, hence $k[x, y]/(y^2 - f(x))$ is integrally closed in $\kappa(C)$ and $U_0 = \mathrm{Spec}(k[x, y]/(y^2 - f(x)))$.

Let $n = \deg(f)$, $m = \lceil \frac{n}{2} \rceil$, $t = x^{-1}$ and $z = t^m y$ in $\kappa(C)$, with the relation $z^2 = t^{2m}f(t^{-1})$. By the same argument, we conclude that $U_\infty = \mathrm{Spec}\left(k[t, z]/\left(z^2 - t^{2m}f(t^{-1})\right)\right)$.

Example 2.3 (Ramification of a hyperelliptic curve). Let $\pi\colon C \to \mathbf{P}^1_k$ be a hyperelliptic curve over k. By Remark 2.2, U_0 and U_∞ form an open covering of C. On U_0, the morphism π is induced by the canonical map

$$k[x] \to k[x,y]/(y^2 - f(x)), \quad x \mapsto x.$$

A point of U_0 corresponds to $(a,b) \in k^2$ such that $b^2 = f(a)$. Since $x-a$ is a uniformizer of $\mathcal{O}_{\mathbf{P}^1_k,a}$, the ramification index at (a,b) is given by $e_{(a,b)} = v_{(a,b)}(x-a)$. Suppose that $f(a) = 0$, hence $b = 0$. By Euclidean division, there exists $h(x) \in k[x]$ such that $f(x) = (x-a)h(x)$, where $h(x)$ is invertible in the local ring $\mathcal{O}_{C,(a,0)}$, as its image in the residue field is $h(a) \neq 0$. In this case, $x - a = h(x)^{-1}y^2$, hence $(x-a, y) = (y)$ and y is a uniformizer of the local ring $\mathcal{O}_{C,(a,0)}$ and $e_{(a,0)} = v_{(a,0)}(x-a) = 2$. Conversely, if $f(a) \neq 0$, then there exists $h(x) \in k[x]$ such that $y^2 - b^2 = f(x) - f(a) = (x-a)h(x)$. As the image of $y + b$ in the residue field is $2b \neq 0$, it is invertible, hence $(x-a, y-b) = (x-a)$ and $x-a$ is a uniformizer of the local ring $\mathcal{O}_{C,(a,b)}$. In particular, $e_{(a,b)} = 1$.

On U_∞, the morphism π is induced by the canonical map

$$k[t] \to k[t,z]/\left(z^2 - t^{2m}f(t^{-1})\right), \quad t \mapsto t.$$

It suffices to consider $t = 0$. Let $n = \deg(f)$ and $F(t) = t^n f(t^{-1})$, where $F(0) \neq 0$. In this case, t is a uniformizer of the local ring $\mathcal{O}_{\mathbf{P}^1_k,\infty} = k[t]_{(t)}$. If n is odd, then $z^2 = tF(t)$, so the unique point above $t = 0$ is $(0,0)$. As $F(t)$ is a unit, $(t,z) = (z)$ and the ramification index at this point is 2. If n is even, then $z^2 = F(t)$ and the ramification index is 1 at both points.

By Theorem 1.9 and the above calculations, we may compute the degree of the ramification divisor R_f as

$$\deg(R_f) = \begin{cases} n+1 & n \text{ odd} \\ n & n \text{ even.} \end{cases}$$

Since $\deg(\pi) = 2$ and $g(\mathbf{P}^1_k) = 0$, the genus of C is computed as $g(C) = \lfloor \frac{n-1}{2} \rfloor$.

Lemma 2.4. *Let $f\colon \mathbf{P}^1_k \to C$ be a dominant morphism of curves over k. If $\mathrm{char}(k) = p$ and f is not generically separable, then there exists a dominant morphism $h\colon \mathbf{P}^1_k \to C$ such that $\deg(h) = p^{-1}\deg(f)$.*

Proof. Since $k(t)$ is a $k(t^p)$-vector space with basis $1, t, \ldots, t^{p-1}$, the kernel of the universal k-derivation $\mathrm{d}\colon k(t) \to \Omega_{k(t)/k}$ is $k(t^p)$. Consider the fundamental exact sequence

$$\Omega_{\kappa(C)/k} \otimes_{\kappa(C)} k(t) \xrightarrow{\eta^* \mathrm{d}f} \Omega_{k(t)/k} \to \Omega_{k(t)/\kappa(C)} \to 0,$$

where $\eta^*\mathrm{d}f(\mathrm{d}a \otimes 1) = \mathrm{d}a$. By Remark 1.2, $\mathrm{d}a = 0$ for every $a \in \kappa(C)$, i.e., $\kappa(C) \subseteq k(t^p)$. This corresponds to a dominant morphism $h\colon \mathbf{P}^1_k \to C$ of degree $\deg(h) = p^{-1}\deg(f)$. □

Theorem 2.5 (Lüroth's theorem). *Let $f\colon \mathbf{P}^1_k \to C$ be a finite dominant morphism of curves over k. Then $C \simeq \mathbf{P}^1_k$.*

Proof. By Lemma 2.4, we may assume without loss of generality that f is generically separable. From Theorem 1.9, the genus of C satisfies

$$-2 = n(2g(C) - 2) + \deg(R_f).$$

Since the ramification divisor R_f is an effective Cartier divisor, $g(C) = 0$. Clearly $C(k) \neq \varnothing$, hence it follows from the classification of genus zero curves that $C \simeq \mathbf{P}^1_k$. □

Proposition 2.6 (Endomorphisms of curves of genus ≥ 2). *Let C be a curve over k such that $g(C) \geq 2$. Every finite, generically separable morphism $f\colon C \to C$ over k is an automorphism.*

Proof. Let $n = \deg(f)$. By Theorem 1.9, the genus of C satisfies

$$0 = (n-1)(2g(C) - 2) + \deg(R_f).$$

Since the ramification divisor R_f is an effective Cartier divisor and $g(C) \geq 2$, we must have $n = 1$. In other words, f induces an isomorphism of function fields, hence f is an isomorphism of curves. □

3. Topological genus

Throughout this section, a *surface* is a compact, connected, oriented topological surface without boundary.

Theorem 3.1 (Classification of closed surfaces)**.** *Every surface S is homeomorphic to a sphere with $g_{\mathrm{top}}(S)$ handles Σ_g, where the unique integer $g_{\mathrm{top}}(S) \geq 0$ is the* topological genus *of S.*

Remark 3.2. In Theorem 3.1, $S \simeq \Sigma_g$ admits a CW-decomposition with a 0-cell, $2g_{\mathrm{top}}(S)$ 1-cell and a 2-cell, given by $4g_{\mathrm{top}}(S)$-gon decomposition. Therefore, the *Euler characteristic* is $\chi(S) = 2 - 2g_{\mathrm{top}}(S)$.

Definition 3.3 (Topological ramification)**.** Let $f\colon X \to Y$ be a map between surfaces.

(a) f is a *branched covering* if it is surjective and locally of the form $z \mapsto z^{e_x}$ at every $x \in X$, where $e_x \geq 1$;
(b) e_x is the *ramification index* of f at x;
(c) $x \in X$ is a *ramification point* and $f(x)$ is a *branch point*, if $e_x > 1$;
(d) the *branch locus* of f is the set of branch points B_f in Y.

Ramification points are isolated, and a proper étale map is a covering map, hence the map $X \setminus f^{-1}(B_f) \to Y \setminus B_f$ is a covering and the following definition is valid:

(e) the *degree* of f is the degree of the covering map $X \setminus f^{-1}(B_f) \to Y \setminus B_f$.

Lemma 3.4. *Let $f\colon X \to Y$ be a branched covering of degree n between surfaces. For every $y \in Y$, $\sum_{x \in f^{-1}(y)} e_x = n$. In particular, $|f^{-1}(y)| = n - \sum_{x \in f^{-1}(y)} (e_x - 1)$.*

Proof. Clearly, $f^{-1}(y)$ is a discrete closed subset of the compact space X, hence finite, i.e., $f^{-1}(y) = \{x_i \mid i \in I\}$, where I is finite. For every $x_i \in f^{-1}(y)$, choose open neighborhoods U_i of x_i and V_i of y such that $f(U_i) \subseteq V_i$ and $f|_{U_i} = z^{e_{x_i}}$. By the compactness of X, there exists some open neighborhood $y \in V \subseteq \bigcap_{i \in I} V_i$ such that $f^{-1}(V) \subseteq \bigcup_{i \in I} U_i$. Let $y' \in V \setminus B_f$. Write $f^{-1}(y') = \bigsqcup_{i \in I} (f^{-1}(y') \cap U_i)$. If y' is sufficiently close to y, then

$$n = |f^{-1}(y')| = \sum_{i \in I} |f^{-1}(y') \cap U_i| = \sum_{i \in I} e_{x_i}.$$

□

Theorem 3.5 (Riemann-hurwitz formula)**.** *Let $f\colon X \to Y$ be a branched covering of degree n between surfaces.*

$$\chi(X) = n\chi(Y) - \sum_{x \in X} (e_x - 1).$$

Proof. Fix a finite triangulation of Y. Via refinement, we may assume that every branch point $b \in B_f$ is a vertex of the triangulation. Denote the sets of vertices, edges and faces by V_Y, E_Y, F_Y respectively. The connected components of the preimages of the open cells under f form a CW-decomposition of X. Since the interior of each face in Y is simply connected, its preimage consists of precisely n copies of 2-cells in X. Similarly, the preimage of the interior of each edge consists of n copies of 1-cells in X. By Lemma 3.4, the number of 0-cells in X is given by

$$n|V_Y| - \sum_{v \in V_Y} \sum_{x \in f^{-1}(v)} (e_x - 1) = n|V_Y| - \sum_{x \in X} (e_x - 1),$$

where the last equality follows, because every branch point is a vertex of the triangulation of Y. We deduce the formula by computing the Euler characteristic of X via the CW-decomposition. □

Remark 3.6 (Topological genus and algebraic genus)*.* Let C be a curve over $\mathbf{C}$. There exists a *complex analytic space* C^{an} associated to C, whose underlying set is the $\mathbf{C}$-points of C. The smoothness and projectivity of C implies that C^{an} is a compact Riemann surface. A morphism $f\colon Y \to X$ of curves over $\mathbf{C}$ induces a proper, finite-to-one holomorphic map $f^{\mathrm{an}}\colon Y^{\mathrm{an}} \to X^{\mathrm{an}}$ between compact Riemann surfaces.

Moreover, if $y \in Y(\mathbf{C})$ and e_y is the ramification index of f at y, the corresponding map f^{an} is given in suitable local holomorphic coordinates by $z \mapsto z^{e_y}$. Furthermore, the degree of f equals the degree of f^{an}.

Choose $t \in \kappa(C)$ transcendental over $\mathbf{C}$. The extension $\mathbf{C}(t) \subseteq \kappa(C)$ corresponds to a finite morphism $\pi\colon C \to \mathbf{P}^1_{\mathbf{C}}$. In fact, $(\mathbf{P}^1_{\mathbf{C}})^{\mathrm{an}} = S^2$, hence $g_{\mathrm{top}}\left((\mathbf{P}^1_{\mathbf{C}})^{\mathrm{an}}\right) = g(\mathbf{P}^1_{\mathbf{C}}) = 0$. By Theorem 1.9 and Theorem 3.5, we conclude that $g_{\mathrm{top}}(C^{\mathrm{an}}) = g(C)$.

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